HSC Mathematics Extension 2

The Nature of Proof

Vu Hung Nguyen

“Mathematics is the language with which God has written the universe.”

— Galileo Galilei

Contents

1 Introduction

1.1 Project Overview

This booklet presents a comprehensive collection of proof problems for HSC Mathematics Extension 2 students. The 37 carefully selected problems cover all essential proof techniques: Direct Proof, Proof by Contradiction, Mathematical Induction, and Proof by Cases. Topics include divisibility, irrationality, modular arithmetic, parity, and logical reasoning.

The collection is divided into two parts:

- **Part 1 (20 problems):** Detailed step-by-step solutions showing complete reasoning, algebraic manipulation, and justification of each step. Each solution is designed to fit within one page, modeling clear and concise proof writing.
- **Part 2 (17 problems):** Strategic hints and solution sketches using flipped hint environments, encouraging independent problem-solving while providing guidance on key steps.

1.2 Target Audience

This booklet is designed for HSC Mathematics Extension 2 students who want to:

- Master fundamental proof techniques required for the HSC examination
- Develop rigorous mathematical reasoning and proof-writing skills
- Build confidence with both standard and non-standard proof problems
- Understand connections between different proof methods
- Practice problems at Easy, Medium, and Hard difficulty levels

Teachers and tutors will also find this collection valuable for:

- Demonstrating worked examples with clear pedagogical progression
- Selecting problems at appropriate difficulty levels for students
- Providing comprehensive coverage of HSC Extension 2 proof topics
- Preparing students for examination standards in mathematical proof

1.3 How to Use This Booklet

For Students:

- Begin with the Proof Primer below to review fundamental concepts and techniques.
- Work through Part 1 problems first, attempting each problem before reading the detailed solution.
- Compare your proof structure and reasoning with the provided solutions.
- Pay attention to the "Takeaways" section after each solution—these highlight key techniques and common pitfalls.
- Move to Part 2 problems, using the upside-down hints only after making a genuine attempt.

- Revisit challenging problems after a few days to reinforce understanding and technique.

For Tutors and Teachers:

- Use Part 1 problems as worked examples in lessons, highlighting proof structure and key techniques.
- Assign Part 2 problems for homework or practice, encouraging students to attempt problems before revealing hints.
- Select problems by proof technique or topic to target specific areas of the syllabus.
- Use the variety of difficulty levels (Easy, Medium, Hard) to differentiate instruction for different student abilities.
- Note Problem 14 (Part 1) as the "Induction Crossover" demonstrating overlap with HSC-Induction collection.

1.4 Proof Primer

Essential Number Theory Concepts

1. Divisibility: $a|b$ (read "a divides b") means there exists an integer k such that $b = ak$.

2. Division Algorithm: For integers a and $b > 0$, there exist unique integers q (quotient) and r (remainder) such that:

$$a = bq + r, \quad 0 \leq r < b$$

This is essential for case-based proofs (e.g., classifying integers as $3k$, $3k + 1$, or $3k + 2$).

3. Modular Arithmetic: $a \equiv b \pmod{m}$ (read "a is congruent to b modulo m") means $m|(a - b)$. Equivalently, a and b have the same remainder when divided by m.

Key Properties:

- If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$
- If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $ac \equiv bd \pmod{m}$
- If $a \equiv b \pmod{m}$, then $a^n \equiv b^n \pmod{m}$ for any positive integer n

4. Parity: An integer is *even* if it can be written as $2k$ for some integer k. An integer is *odd* if it can be written as $2k + 1$ for some integer k.

5. Prime Numbers: An integer $p > 1$ is prime if its only positive divisors are 1 and p. Key property: If prime p divides ab, then $p|a$ or $p|b$.

Essential Proof Techniques

1. Direct Proof: Start with given information and use logical steps to reach the desired conclusion.

- Example: To prove $a|b$ and $b|c$ implies $a|c$, write $b = ak_1$ and $c = bk_2$, then $c = (ak_1)k_2 = a(k_1k_2)$.

2. Proof by Contradiction (Reductio ad Absurdum):

- Assume the negation of what you want to prove
- Use logical reasoning to derive a statement that contradicts a known fact, the hypothesis, or a mathematical truth

- Conclude that the assumption must be false, hence the original statement is true
- Example: Proving $\sqrt{2}$ is irrational by assuming $\sqrt{2} = \frac{p}{q}$ and deriving a contradiction

3. Mathematical Induction: To prove $P(n)$ is true for all integers $n \geq n_0$:

- *Base Case:* Prove $P(n_0)$ is true
- *Inductive Hypothesis:* Assume $P(k)$ is true for some arbitrary $k \geq n_0$
- *Inductive Step:* Prove $P(k+1)$ is true using the assumption that $P(k)$ is true
- *Conclusion:* By the principle of mathematical induction, $P(n)$ is true for all $n \geq n_0$

4. Proof by Cases: When the hypothesis can be divided into distinct cases:

- Identify all possible cases (ensure they are exhaustive and mutually exclusive)
- Prove the statement for each case separately
- Conclude that the statement holds in all cases
- Example: Proving $n^2 \equiv 0$ or $1 \pmod 4$ by checking cases n even and n odd

5. Biconditional Proof (If and Only If): To prove "P if and only if Q" (written $P \iff Q$):

- Prove the forward direction: $P \implies Q$
- Prove the reverse direction: $Q \implies P$
- Both directions must be proven independently

6. Inequality Proofs: Common strategies for proving inequalities:

- *Algebraic manipulation:* Transform the inequality to an obviously true statement (show $A \geq B$ by proving $A - B \geq 0$)
- *AM-GM Inequality:* For non-negative numbers: $\frac{a_1 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 \cdots a_n}$
- *Cauchy-Schwarz:* $(a_1b_1 + \cdots + a_nb_n)^2 \leq (a_1^2 + \cdots + a_n^2)(b_1^2 + \cdots + b_n^2)$
- *Induction:* For inequalities involving n, establish base case and inductive step

Induction Variants

Beyond the standard induction framework, several specialized variants frequently appear:

Induction for Series: Proving closed-form formulas for sums (e.g., $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$). In the inductive step, add the $(k+1)$-th term to both sides of $P(k)$ and simplify algebraically.

Induction for Divisibility: Proving statements like "$n|f(n)$ for all $n \geq n_0$". Use the inductive hypothesis to express $f(k+1)$ in terms of $f(k)$ and factor to show divisibility.

Induction for Inequalities: Proving statements like "$2^n > n^2$ for $n \geq 5$". The inductive step requires showing $f(k+1) > g(k+1)$ using the assumption $f(k) > g(k)$, often with careful estimation.

Induction for Recursive Sequences: For recursively defined sequences $a_{n+1} = f(a_n)$, prove properties by expressing a_{k+1} using the recurrence and applying the inductive hypothesis to a_k.

Strong Induction: Assume $P(n_0), P(n_0+1), \ldots, P(k)$ all true, then prove $P(k+1)$. Essential when the $(k+1)$-th case depends on multiple previous cases.

Common Proof Strategies

Contrapositive: To prove $P \implies Q$, instead prove $\neg Q \implies \neg P$ (logically equivalent).

Consecutive Integers: Among k consecutive integers, exactly one is divisible by k. Among 2 consecutive integers, exactly one is even.

Irrationality Proofs: Standard template for proving $\sqrt{n}$ irrational (where n is not a perfect square):

1. Assume $\sqrt{n} = \frac{p}{q}$ in lowest terms ($\gcd(p, q) = 1$)
2. Square both sides: $n = \frac{p^2}{q^2}$, so $p^2 = nq^2$
3. Show $n|p^2$ implies $n|p$ (using prime factorization if n is prime)
4. Substitute $p = nk$, derive $q^2 = nk^2$, showing $n|q$
5. Contradict $\gcd(p, q) = 1$

1.5 The Language of Logic: Reading Quantifier Statements

A key skill in proof writing is translating symbolic logic into plain English and back. The two most important quantifiers are:

- $\forall$ — "for all" / "for every" / "for each" (the *universal* quantifier)
- $\exists$ — "there exists" / "there is at least one" (the *existential* quantifier)

The examples below show both a *literal, symbol-by-symbol* translation and a *natural English* paraphrase. Being able to move fluently between these two registers is essential for understanding what a statement actually claims before you attempt to prove (or disprove) it.

Statement 1 $\forall x \in \mathbb{R},\ x^2 \geq 0$

- *Literal:* For all x in the real numbers, x squared is greater than or equal to zero.
- *Natural English:* The square of any real number is always non-negative.

Statement 2 $\forall n \in \mathbb{Z},\ \exists m \in \mathbb{Z},\ m = n + 5$

- *Literal:* For every integer n, there exists an integer m such that m equals n plus 5.
- *Natural English:* Given any integer, you can always find another integer that is exactly 5 greater than it. *(True.)*

Statement 3 $\exists a \in \mathbb{R},\ \forall x \in \mathbb{R},\ ax = x$

- *Literal:* There exists a real number a such that for all real numbers x, a times x equals x.
- *Natural English:* There is a specific real number that, when multiplied by any real number, leaves that number unchanged. *(True; that number is $a = 1$, the multiplicative identity.)*

Statement 4 $\exists m \in \mathbb{Z},\ \forall n \in \mathbb{Z},\ m = n + 5$

- *Literal:* There exists an integer m such that for every integer n, m equals n plus 5.

- *Natural English:* There is a single "magic" integer that is simultaneously exactly 5 greater than *every* other integer. *(False.)*

Key Takeaway: Order of quantifiers matters.

Statements 2 and 4 use *identical symbols* arranged in a slightly different order, yet they have opposite truth values. This is one of the most common traps in the language of proof:

- **Statement 2** ($\forall$ first, then $\exists$): the choice of m is allowed to *depend* on n. You name an n, and only then do you pick a matching m. This makes the statement true.
- **Statement 4** ($\exists$ first, then $\forall$): the m must be chosen *before* n is known. It claims one fixed constant m works for every single n simultaneously — which is impossible.

Quantifier Rule of Thumb

Swapping the order of $\forall$ and $\exists$ completely changes the mathematical meaning of a sentence. Always read quantifiers left-to-right and ask: *who gets to choose first?*

2 Part 1: Problems with Detailed Solutions

Easy Problems (6 problems)

Problem 2.1: Comparing Radical Sums by Contradiction

Prove by contradiction that $\sqrt{3}+\sqrt{5} > \sqrt{11}$.

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.1

Proof by Contradiction

Step 1: Assume the negation

Assume, for the sake of contradiction, that the statement is false. That is, assume:

$$\sqrt{3}+\sqrt{5} \leq \sqrt{11}$$

Step 2: Square both sides

Since all terms are positive, we can square both sides without changing the inequality:

$$\begin{aligned}(\sqrt{3}+\sqrt{5})^2 &\leq (\sqrt{11})^2\\ 3+2\sqrt{3}\cdot\sqrt{5}+5 &\leq 11\\ 8+2\sqrt{15} &\leq 11\\ 2\sqrt{15} &\leq 3\end{aligned}$$

Step 3: Square again

Both sides are still positive, so square again:

$$\begin{aligned}(2\sqrt{15})^2 &\leq 3^2\\ 4\cdot 15 &\leq 9\\ 60 &\leq 9\end{aligned}$$

Step 4: Establish contradiction

The statement $60 \leq 9$ is clearly false.

This contradiction arose from our assumption that $\sqrt{3}+\sqrt{5} \leq \sqrt{11}$.

Therefore, our assumption must be false, and we conclude:

$$\sqrt{3}+\sqrt{5} > \sqrt{11}$$

□

Takeaways 2.1

- **Proof by Contradiction Structure:** Assume the negation of what you want to prove, derive a logical impossibility, conclude original statement must be true
- **Squaring Inequalities:** When both sides are positive, squaring preserves the inequality direction
- **Algebraic Manipulation:** Expand $(\sqrt{3}+\sqrt{5})^2$ carefully: $(a+b)^2 = a^2+2ab+b^2$
- **Clear Contradictions:** A numerical impossibility like $60 \leq 9$ is an immediate and decisive contradiction

Problem 2.2: No Positive Integer Difference of One

Show that there are no positive integers x and y such that $x^2 - y^2 = 1$.

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.2

Proof by Factorization and Case Analysis

Step 1: Factor the left side

We factor $x^2 - y^2$ as a difference of squares:

$$x^2 - y^2 = (x-y)(x+y) = 1$$

Step 2: Analyze integer factor pairs

Since x and y are positive integers, both $(x-y)$ and $(x+y)$ are integers. We need their product to equal 1.

The only ways to write 1 as a product of integers are:

- $1 = 1 \times 1$
- $1 = (-1) \times (-1)$

Step 3: Case 1 - Both factors equal 1

If $x - y = 1$ and $x + y = 1$, adding these equations gives:

$$2x = 2 \implies x = 1$$

Subtracting: $2y = 0 \implies y = 0$

But $y = 0$ contradicts the requirement that y is a positive integer.

Step 4: Case 2 - Both factors equal -1

If $x - y = -1$ and $x + y = -1$, adding these equations gives:

$$2x = -2 \implies x = -1$$

But $x = -1$ contradicts the requirement that x is a positive integer.

Conclusion

All possible integer factorizations of 1 lead to violations of the positive integer requirement.

Therefore, there are no positive integers x and y satisfying $x^2 - y^2 = 1$.

□

Takeaways 2.2

- **Factorization Strategy:** Recognize $x^2 - y^2 = (x-y)(x+y)$ as difference of squares
- **Integer Factor Analysis:** For product $(x-y)(x+y) = 1$, only factor pairs are $(1,1)$ and $(-1,-1)$
- **Systematic Case Checking:** Solve simultaneous equations $x-y = a$ and $x+y = b$ for each factor pair (a,b)
- **Constraint Verification:** Always check solutions against domain restrictions (here: positive integers)

Problem 2.3: Irrationality of sqrt(23)

Prove that $\sqrt{23}$ is irrational.

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.3

Proof by Contradiction

Step 1: Assume the negation

Assume, for contradiction, that $\sqrt{23}$ is rational. Then it can be written as:

$$\sqrt{23} = \frac{p}{q}$$

where $p, q \in \mathbb{Z}$, $q \neq 0$, and $\gcd(p, q) = 1$ (the fraction is in lowest terms).

Step 2: Square both sides

$$23 = \frac{p^2}{q^2}$$
$$p^2 = 23q^2$$

Step 3: Deduce divisibility

From $p^2 = 23q^2$, we see that p^2 is divisible by 23.

Since 23 is prime, if $23 \mid p^2$, then $23 \mid p$ (by fundamental theorem of arithmetic).

Step 4: Substitute and derive contradiction

Since $23 \mid p$, write $p = 23k$ for some integer k. Substituting into $p^2 = 23q^2$:

$$(23k)^2 = 23q^2$$
$$529k^2 = 23q^2$$
$$23k^2 = q^2$$

This shows q^2 is divisible by 23. Since 23 is prime, $23 \mid q$.

Step 5: Establish contradiction

We have shown that both p and q are divisible by 23.

This contradicts our assumption that $\gcd(p, q) = 1$.

Conclusion

Therefore, $\sqrt{23}$ cannot be expressed as a fraction of integers.

Hence, $\sqrt{23}$ is irrational.

□

Takeaways 2.3

- **Standard Irrationality Template:** Assume rational form $\frac{p}{q}$ in lowest terms, square to get $p^2 = nq^2$, show both p and q divisible by prime factor, contradict lowest terms assumption
- **Prime Divisibility:** If prime p divides a^2, then p divides a (key property from fundamental theorem of arithmetic)
- **Lowest Terms:** Always start with $\gcd(p, q) = 1$ to enable the contradiction via common factors
- **Generalization:** This method proves $\sqrt{p}$ irrational for any prime p

Problem 2.4: Irrationality of Non-Perfect-Square Roots

Let a be a positive integer. If a is not a perfect square, prove that $\sqrt{a}$ is irrational.

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.4

Proof by Contradiction

Step 1: Assume the negation

Assume, for contradiction, that $\sqrt{a}$ is rational. Then we can write

$$\sqrt{a} = \frac{p}{q}$$

where $p, q \in \mathbb{Z}$, $q \neq 0$, and $\gcd(p, q) = 1$.

Step 2: Square both sides

$$a = \frac{p^2}{q^2}$$
$$p^2 = aq^2$$

Step 3: Compare prime factorizations

By the Fundamental Theorem of Arithmetic, every positive integer has a unique prime factorization. When an integer is squared, every prime exponent in its factorization becomes even. So:

- in p^2, every prime appears with an even exponent;
- in q^2, every prime appears with an even exponent.

Now the equation

$$p^2 = aq^2$$

shows that aq^2 must also have only even prime exponents.

Since q^2 already contributes only even exponents, this is possible only if every prime appearing in a also has an even exponent.

Step 4: Derive the contradiction

But if every prime in the factorization of a has an even exponent, then a is a perfect square.

This contradicts the hypothesis that a is *not* a perfect square.

Conclusion

Therefore, our assumption was false, and $\sqrt{a}$ must be irrational.

□

Takeaways 2.4

- **Parity of Prime Exponents:** A perfect square has only even exponents in its prime factorization
- **Why the Method Works:** Squaring forces all prime exponents in p^2 and q^2 to be even, so the same must be true for a
- **Generalization:** This extends proofs like "$\sqrt{2}$ is irrational" or "$\sqrt{23}$ is irrational" to any positive integer that is not a square
- **Key Tool:** The argument relies on the Fundamental Theorem of Arithmetic and uniqueness of prime factorization

Problem 2.5: Contrapositive of Multiples of 6

Consider the statement: "For all integers n, if n is a multiple of 6, then n is a multiple of 2."
Which of the following is the contrapositive of this statement?

A. There exists an integer n such that n is a multiple of 6 and not a multiple of 2.

B. There exists an integer n such that n is a multiple of 2 and not a multiple of 6.

C. For all integers n, if n is not a multiple of 2, then n is not a multiple of 6.

D. For all integers n, if n is not a multiple of 6, then n is not a multiple of 2.

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.5

Logical Analysis of Contrapositive
Step 1: Identify the logical structure
The original statement has the form:

$$\forall n \in \mathbb{Z}, \quad P(n) \implies Q(n)$$

Where:

- $P(n)$: "n is a multiple of 6"
- $Q(n)$: "n is a multiple of 2"

Step 2: Apply contrapositive definition
The contrapositive of $P \implies Q$ is $\neg Q \implies \neg P$.
Key: The universal quantifier ("for all") remains unchanged.
Step 3: Negate each part

- $\neg Q(n)$: "n is NOT a multiple of 2"
- $\neg P(n)$: "n is NOT a multiple of 6"

Step 4: Construct the contrapositive

$$\forall n \in \mathbb{Z}, \quad \neg Q(n) \implies \neg P(n)$$

In words: "For all integers n, if n is not a multiple of 2, then n is not a multiple of 6."
This matches **Option C**.
Analysis of incorrect options:

- **Option A:** Negation of original ($\exists n : P(n) \land \neg Q(n)$), not contrapositive
- **Option B:** Negation of converse
- **Option D:** Inverse ($\neg P \implies \neg Q$), not contrapositive

Answer: $\boxed{\text{C}}$

Takeaways 2.5

- **Contrapositive Form:** $P \implies Q$ has contrapositive $\neg Q \implies \neg P$ (swap and negate both parts)
- **Logical Equivalence:** A statement and its contrapositive are logically equivalent (same truth value)
- **Quantifiers Unchanged:** Universal quantifier ("for all") stays when forming contrapositive
- **Common Errors:** Inverse ($\neg P \implies \neg Q$) and converse ($Q \implies P$) are NOT equivalent to original

Problem 2.6: Divisibility of n Squared Minus One by Three

Prove that $n^2 - 1$ is divisible by 3 if n is not a multiple of 3.

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.6

Proof by Cases (Modular Arithmetic)
Step 1: Set up the cases
If n is not a multiple of 3, then $n \not\equiv 0 \pmod 3$.
By the division algorithm, n must satisfy one of:

- $n \equiv 1 \pmod 3$, or
- $n \equiv 2 \pmod 3$

We will prove $n^2 - 1 \equiv 0 \pmod 3$ in both cases.
Step 2: Case 1 - $n \equiv 1 \pmod 3$

$$\begin{aligned} n^2 - 1 &\equiv (1)^2 - 1 \pmod 3 \\ &\equiv 1 - 1 \pmod 3 \\ &\equiv 0 \pmod 3 \end{aligned}$$

Therefore, $3 \mid (n^2 - 1)$ in this case.
Step 3: Case 2 - $n \equiv 2 \pmod 3$

$$\begin{aligned} n^2 - 1 &\equiv (2)^2 - 1 \pmod 3 \\ &\equiv 4 - 1 \pmod 3 \\ &\equiv 3 \pmod 3 \\ &\equiv 0 \pmod 3 \end{aligned}$$

Therefore, $3 \mid (n^2 - 1)$ in this case as well.
Conclusion
Since $n^2 - 1 \equiv 0 \pmod 3$ in all possible cases where $3 \nmid n$, we conclude that $n^2 - 1$ is divisible by 3 whenever n is not a multiple of 3.

□

Takeaways 2.6

- **Proof by Cases Setup:** If $n \not\equiv 0 \pmod m$, check all other residue classes modulo m
- **Modular Arithmetic:** Use $a \equiv b \pmod m$ to mean $m \mid (a - b)$, or equivalently, a and b have same remainder when divided by m
- **Complete Case Coverage:** For modulo 3, only residues are 0, 1, 2; excluding 0 leaves exactly 2 cases to check
- **Calculation Technique:** Substitute residue values directly: if $n \equiv 2 \pmod 3$, then $n^2 \equiv 4 \equiv 1 \pmod 3$

Medium Problems (5 problems)

Problem 2.7: Odd Squares and Divisibility by 8

Prove that if a is any odd integer, then $a^2 - 1$ is divisible by 8.

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.7

Direct Proof

Step 1: Express a as an odd integer

Since a is odd, we can write:

$$a = 2k + 1$$

for some integer k.

Step 2: Expand $a^2 - 1$

$$\begin{aligned} a^2 - 1 &= (2k+1)^2 - 1 \\ &= 4k^2 + 4k + 1 - 1 \\ &= 4k^2 + 4k \\ &= 4k(k+1) \end{aligned}$$

Step 3: Analyze $k(k+1)$

Note that k and $(k+1)$ are consecutive integers.

Therefore, one of them must be even, which means their product $k(k+1)$ is divisible by 2.

Write $k(k+1) = 2m$ for some integer m.

Step 4: Substitute and conclude

$$\begin{aligned} a^2 - 1 &= 4 \cdot k(k+1) \\ &= 4 \cdot 2m \\ &= 8m \end{aligned}$$

Since $a^2 - 1 = 8m$, we conclude that $a^2 - 1$ is divisible by 8.

□

Takeaways 2.7

- **Odd Integer Form:** Any odd integer can be written as $2k + 1$ for some integer k
- **Consecutive Integer Property:** Product of consecutive integers $k(k+1)$ is always even (one must be even)
- **Factor Extraction:** From $4k(k+1) = 4 \cdot 2m = 8m$, directly see divisibility by 8
- **Alternative View:** Can also factor $a^2 - 1 = (a-1)(a+1)$, both even for odd a, with one divisible by 4

Problem 2.8: Quadratic Residues Modulo 5

(a) Given a is integral and not divisible by 5, prove the remainder when a^2 is divided by 5 is either 1 or 4.

(b) Hence, given that a, b are integral and not divisible by 5, prove that $a^4 - b^4$ is divisible by 5.

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.8

Part (a): Proof by Cases
Since a is not divisible by 5, we have $a \not\equiv 0 \pmod 5$.
By the division algorithm, a must be congruent to 1, 2, 3, or 4 modulo 5.
We check each case:
Case 1: $a \equiv 1 \pmod 5$

$$a^2 \equiv 1^2 \equiv 1 \pmod 5$$

Case 2: $a \equiv 2 \pmod 5$

$$a^2 \equiv 2^2 \equiv 4 \pmod 5$$

Case 3: $a \equiv 3 \pmod 5$

$$a^2 \equiv 3^2 \equiv 9 \equiv 4 \pmod 5$$

Case 4: $a \equiv 4 \pmod 5$

$$a^2 \equiv 4^2 \equiv 16 \equiv 1 \pmod 5$$

In all cases, $a^2 \equiv 1 \text{ or } 4 \pmod 5$.
Therefore, the remainder when a^2 is divided by 5 is either 1 or 4. □

Part (b): Using Part (a)
From part (a), for any integer x not divisible by 5: $x^2 \equiv 1 \text{ or } 4 \pmod 5$.
Consider $a^4 = (a^2)^2$:

- If $a^2 \equiv 1 \pmod 5$, then $(a^2)^2 \equiv 1^2 \equiv 1 \pmod 5$
- If $a^2 \equiv 4 \pmod 5$, then $(a^2)^2 \equiv 4^2 \equiv 16 \equiv 1 \pmod 5$

Thus $a^4 \equiv 1 \pmod 5$ for any integer a not divisible by 5.
Similarly, $b^4 \equiv 1 \pmod 5$.
Therefore:

$$\begin{aligned} a^4 - b^4 &\equiv 1 - 1 \pmod 5 \\ &\equiv 0 \pmod 5 \end{aligned}$$

Hence $5 \mid (a^4 - b^4)$. □

Takeaways 2.8

- **Systematic Case Analysis:** For $5 \nmid a$, check all residues $a \equiv 1, 2, 3, 4 \pmod 5$ systematically
- **Squaring Congruences:** If $a \equiv b \pmod m$, then $a^2 \equiv b^2 \pmod m$
- **"Hence" Strategy:** Part (b) builds directly on part (a)'s result; apply it twice to get $a^4 \equiv b^4 \equiv 1 \pmod 5$
- **Fermat's Little Theorem Preview:** Result $a^4 \equiv 1 \pmod 5$ for $\gcd(a, 5) = 1$ is special case of FLT

Problem 2.9: Contradiction with a + b ¡= 5

Prove by contradiction that if a, b are integers and $a + b \leq 5$, then $a \leq 2$ or $b \leq 2$.

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.9

Proof by Contradiction
Step 1: Assume the negation of the conclusion
We want to prove: If $a + b \leq 5$, then $a \leq 2$ or $b \leq 2$.
Assume the conclusion is false. The negation of "$a \leq 2$ or $b \leq 2$" is:

$$a > 2 \quad \text{AND} \quad b > 2$$

Step 2: Apply the integer constraint
Since a and b are integers with $a > 2$ and $b > 2$, we must have:

$$a \geq 3 \quad \text{and} \quad b \geq 3$$

(The smallest integer greater than 2 is 3.)
Step 3: Derive a contradiction
Adding these inequalities:

$$a + b \geq 3 + 3$$
$$a + b \geq 6$$

But this contradicts our hypothesis that $a + b \leq 5$.
We cannot have both $a + b \geq 6$ and $a + b \leq 5$ simultaneously.
Conclusion
Since assuming the negation of the conclusion leads to a contradiction, the conclusion must be true.
Therefore, if a, b are integers with $a + b \leq 5$, then $a \leq 2$ or $b \leq 2$.

□

Takeaways 2.9

- **Negating "Or" Statements:** The negation of "P or Q" is "not P AND not Q"
- **Integer Constraints:** For integers, $a > 2$ implies $a \geq 3$ (no integers strictly between 2 and 3)
- **Contradiction Structure:** Derive statement that directly contradicts a hypothesis (here: $a + b \geq 6$ vs $a + b \leq 5$)
- **Logical Form:** Statement has form $(P \implies Q \vee R)$; negate conclusion to get $\neg Q \wedge \neg R$

Problem 2.10: Last-Two-Digits Test for 4

Prove that a number is divisible by 4 if and only if the last two digits form a number divisible by 4.

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.10

Biconditional Proof (If and Only If)
This requires proving both directions:
Forward Direction ($\implies$): If N is divisible by 4, then its last two digits form a number divisible by 4.
Reverse Direction ($\impliedby$): If the last two digits form a number divisible by 4, then N is divisible by 4.

Setup: Express any integer N as:

$$N = 100A + L$$

where A is the number formed by all digits except the last two, and L is the two-digit number formed by the last two digits ($0 \leq L \leq 99$).

Forward Direction Proof:
Assume $4 \mid N$, so $N \equiv 0 \pmod 4$.
Then:

$$100A + L \equiv 0 \pmod 4$$

Since $100 = 4 \times 25$, we have $100A \equiv 0 \pmod 4$.
Therefore:

$$\begin{aligned} 0 + L &\equiv 0 \pmod 4 \\ L &\equiv 0 \pmod 4 \end{aligned}$$

Thus $4 \mid L$. $\square$

Reverse Direction Proof:
Assume $4 \mid L$, so $L \equiv 0 \pmod 4$.
Since $100 = 4 \times 25$, we have $100A \equiv 0 \pmod 4$.
Therefore:

$$\begin{aligned} N = 100A + L &\equiv 0 + 0 \pmod 4 \\ N &\equiv 0 \pmod 4 \end{aligned}$$

Thus $4 \mid N$. $\square$

Conclusion:
Since both directions are proven, we conclude:

$$N \text{ is divisible by } 4 \iff \text{last two digits of } N \text{ divisible by } 4$$

Takeaways 2.10

- **Iff Proof Structure:** Must prove both ($\implies$) and ($\impliedby$) directions independently
- **Digit Representation:** Writing $N = 100A + L$ separates last two digits for modular analysis
- **Key Observation:** Since $100 \equiv 0 \pmod 4$, divisibility of N by 4 depends only on last two digits
- **Generalization:** Same technique proves divisibility rules for powers of 2 (e.g., rule for 8 uses last three digits)

Problem 2.11: Parity of a Difference of Squares

Prove that if m, n are integers, then $m^2 - n^2$ is even if and only if at least one of $(m+n)$ and $(m-n)$ is even.

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.11

Biconditional Proof
Step 1: Factor the expression

$$m^2 - n^2 = (m-n)(m+n)$$

This factorization will be used in both directions.

Forward Direction ($\Longrightarrow$): If $m^2 - n^2$ is even, then at least one of $(m+n)$ or $(m-n)$ is even.

Proof. Since $m^2 - n^2$ is even, the product $(m-n)(m+n)$ is even.
A product of two integers is even if and only if at least one factor is even.
Therefore, at least one of $(m-n)$ or $(m+n)$ must be even. □ □

Reverse Direction ($\Longleftarrow$): If at least one of $(m+n)$ or $(m-n)$ is even, then $m^2 - n^2$ is even.

Proof. We consider two cases:
Case 1: $(m+n)$ is even.
Write $(m+n) = 2k$ for some integer k.
Then:

$$m^2 - n^2 = (m-n)(m+n) = (m-n)(2k) = 2k(m-n)$$

Since $k(m-n)$ is an integer, $m^2 - n^2$ is even.
Case 2: $(m-n)$ is even.
Write $(m-n) = 2j$ for some integer j.
Then:

$$m^2 - n^2 = (m-n)(m+n) = (2j)(m+n) = 2j(m+n)$$

Since $j(m+n)$ is an integer, $m^2 - n^2$ is even.
In both cases, $m^2 - n^2$ is even. □ □

Conclusion:
Both directions proven, therefore:

$$m^2 - n^2 \text{ is even} \iff \text{at least one of } (m+n), (m-n) \text{ is even}$$

Takeaways 2.11

- **Factorization First:** Factoring $m^2 - n^2 = (m-n)(m+n)$ immediately connects to parity of factors
- **Product Parity:** Product even $\iff$ at least one factor even (fundamental parity property)
- **Case Analysis:** Reverse direction handles two cases (either factor even) separately
- **Note on Parity:** Actually, $(m+n)$ and $(m-n)$ always have same parity (both even or both odd), so "at least one even" is equivalent to "both even"

Hard Problems (9 problems)

Problem 2.12: Impossible Exponential Diophantine Equation

Explain why there is no integer n such that $(n+1)^{41} - 79n^{40} = 2$.

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.12

Proof by Modular Analysis and Case Checking

Step 1: Test $n = 0$

If $n = 0$:

$$(0+1)^{41} - 79(0)^{40} = 1 - 0 = 1 \neq 2$$

So $n = 0$ is not a solution.

Step 2: Analyze for $n \neq 0$ using modular arithmetic

For $n \neq 0$, consider the equation modulo n:

By the Binomial Theorem:

$$(n+1)^{41} = \sum_{k=0}^{41} \binom{41}{k} n^k \cdot 1^{41-k}$$

All terms contain n except the last term ($k = 0$):

$$(n+1)^{41} \equiv 1 \pmod{n}$$

The term $-79n^{40}$ is clearly divisible by n:

$$-79n^{40} \equiv 0 \pmod{n}$$

Therefore:

$$\begin{aligned} (n+1)^{41} - 79n^{40} &\equiv 2 \pmod{n} \\ 1 - 0 &\equiv 2 \pmod{n} \\ 1 &\equiv 2 \pmod{n} \end{aligned}$$

Step 3: Interpret the congruence

$1 \equiv 2 \pmod{n}$ means $n \mid (1-2)$, so $n \mid (-1)$.

Therefore $n \in \{-1, 1\}$.

Step 4: Test candidate values

For $n = 1$:

$$(1+1)^{41} - 79(1)^{40} = 2^{41} - 79 = 2,199,023,255,552 - 79 \neq 2$$

For $n = -1$:

$$(-1+1)^{41} - 79(-1)^{40} = 0 - 79(1) = -79 \neq 2$$

Conclusion

All possible integer values of n have been checked and none satisfy the equation.

Therefore, there is no integer n such that $(n+1)^{41} - 79n^{40} = 2$.

□

Takeaways 2.12

- **Modular Arithmetic Strategy:** Working mod n can dramatically constrain possible solutions
- **Binomial Expansion Mod n:** $(n+1)^k \equiv 1 \pmod{n}$ since all terms except constant contain n
- **Divisor Analysis:** $1 \equiv 2 \pmod{n}$ means $n \mid (-1)$, giving only $n \in \{-1, 1\}$
- **Exhaustive Checking:** After constraining to finitely many cases, verify each directly

Problem 2.13: Nested Radicals and a Cosine Formula

The numbers a_n, for integers $n \geq 1$, are defined as:

$$a_1 = \sqrt{2}$$
$$a_2 = \sqrt{2+\sqrt{2}}$$
$$a_3 = \sqrt{2+\sqrt{2+\sqrt{2}}}, \text{ and so on.}$$

These numbers satisfy the relation $a_{n+1}^2 = 2 + a_n$, for $n \geq 1$. (Do NOT prove this.) Use mathematical induction to prove that $a_n = 2\cos\left(\frac{\pi}{2^{n+1}}\right)$ for all integers $n \geq 1$.

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.13

Proof by Mathematical Induction
Base Case $(n = 1)$:
LHS: $a_1 = \sqrt{2}$ (given)
RHS: $2\cos\left(\frac{\pi}{2^{1+1}}\right) = 2\cos\left(\frac{\pi}{4}\right) = 2 \cdot \frac{\sqrt{2}}{2} = \sqrt{2}$
Since LHS = RHS, the statement holds for $n = 1$. ✓

Inductive Hypothesis:
Assume the statement is true for $n = k$, where k is a positive integer:

$$a_k = 2\cos\left(\frac{\pi}{2^{k+1}}\right)$$

Inductive Step:
We must prove the statement for $n = k + 1$:

$$a_{k+1} = 2\cos\left(\frac{\pi}{2^{k+2}}\right)$$

Using the recurrence relation $a_{k+1}^2 = 2 + a_k$ and the inductive hypothesis:

$$\begin{aligned} a_{k+1}^2 &= 2 + 2\cos\left(\frac{\pi}{2^{k+1}}\right) \\ &= 2\left[1 + \cos\left(\frac{\pi}{2^{k+1}}\right)\right] \end{aligned}$$

Apply the double angle identity: $1 + \cos(2\theta) = 2\cos^2(\theta)$.
Let $2\theta = \frac{\pi}{2^{k+1}}$, so $\theta = \frac{\pi}{2^{k+2}}$:

$$\begin{aligned} a_{k+1}^2 &= 2 \cdot 2\cos^2\left(\frac{\pi}{2^{k+2}}\right) \\ &= 4\cos^2\left(\frac{\pi}{2^{k+2}}\right) \end{aligned}$$

Taking square roots:

$$a_{k+1} = \pm 2\cos\left(\frac{\pi}{2^{k+2}}\right)$$

Since $n \geq 1$, we have $0 < \frac{\pi}{2^{k+2}} < \frac{\pi}{2}$, so $\cos\left(\frac{\pi}{2^{k+2}}\right) > 0$.
Also, a_n is defined as nested square roots of positive numbers, so $a_{k+1} > 0$.
Therefore:

$$a_{k+1} = 2\cos\left(\frac{\pi}{2^{k+2}}\right)$$

This completes the inductive step. ✓

Conclusion:
By mathematical induction, $a_n = 2\cos\left(\frac{\pi}{2^{n+1}}\right)$ for all integers $n \geq 1$.

□

Takeaways 2.13

- **Induction with Recurrence:** Use given recurrence relation in inductive step to relate a_{k+1} to a_k
- **Double Angle Formula:** Identity $1 + \cos(2\theta) = 2\cos^2(\theta)$ is key to converting sum to square
- **Sign Consideration:** Must justify taking positive square root using domain/range analysis
- **Angle Halving:** Pattern shows a_n relates to cos of successively halved angles $(\pi/4, \pi/8, \pi/16, \ldots)$

Problem 2.14: Continued Fraction for 1 + sqrt(2)

Let $a = 1 + \sqrt{2}$.

(i) Show that $a^2 - 2a - 1 = 0$.

(ii) Define $x_1 = 2$ and $x_{n+1} = 2 + \frac{1}{x_n}$ for $n \geq 1$. Assuming $x_n \to L$ for some positive limit L, prove that $L = a$.

(iii) Show that $1 + \frac{1}{a} = \sqrt{2}$.

(iv) Hence deduce that

$$\sqrt{2} = 1 + \cfrac{1}{2 + \frac{1}{2 + \frac{1}{2 + \dots}}}.$$

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.14

(i) Since $a = 1 + \sqrt{2}$, we have $a - 1 = \sqrt{2}$. Squaring:

$$(a-1)^2 = 2 \implies a^2 - 2a + 1 = 2 \implies a^2 - 2a - 1 = 0.$$

(ii) If $x_n \to L$, then also $x_{n+1} \to L$. Taking limits in

$$x_{n+1} = 2 + \frac{1}{x_n}$$

gives

$$L = 2 + \frac{1}{L}.$$

So

$$L^2 - 2L - 1 = 0.$$

By part (i), a satisfies the same quadratic. The roots are $1 \pm \sqrt{2}$, and since $L > 0$, we get

$$L = 1 + \sqrt{2} = a.$$

(iii) Using $a = 1 + \sqrt{2}$,

$$1 + \frac{1}{a} = 1 + \frac{1}{1+\sqrt{2}} = 1 + \frac{1-\sqrt{2}}{(1+\sqrt{2})(1-\sqrt{2})} = 1 - (1 - \sqrt{2}) = \sqrt{2}.$$

(iv) The recursion generates

$$x_1 = 2, \quad x_2 = 2 + \frac{1}{2}, \quad x_3 = 2 + \frac{1}{2 + \frac{1}{2}}, \ \dots$$

Hence its limit is exactly

$$2 + \cfrac{1}{2 + \frac{1}{2 + \dots}}.$$

By part (ii), this limit is a, so part (iii) gives

$$\sqrt{2} = 1 + \frac{1}{a} = 1 + \cfrac{1}{2 + \frac{1}{2 + \frac{1}{2 + \dots}}}.$$

□

Takeaways 2.14

- **Limit Method:** Infinite continued fractions are handled by defining finite approximations and taking a limit
- **Fixed Point Idea:** A convergent recursive sequence often leads to an equation satisfied by its limit
- **Quadratic Link:** The continued fraction for $\sqrt{2}$ comes from solving a simple quadratic
- **Representation:** This gives a classic infinite continued fraction expansion for $\sqrt{2}$

Problem 2.15: Irrationality of Log Base n of n Plus One

Prove that for any integer $n > 1$, $\log_n(n+1)$ is irrational.

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.15

Proof by Contradiction

Step 1: Assume the negation

Assume, for contradiction, that $\log_n(n+1)$ is rational for some integer $n > 1$.
Then we can write:

$$\log_n(n+1) = \frac{p}{q}$$

where p, q are positive integers.

Step 2: Convert to exponential form

By definition of logarithm:

$$n^{p/q} = n + 1$$

Raising both sides to the power q:

$$n^p = (n+1)^q$$

Step 3: Analyze modulo n

Left side: $n^p \equiv 0 \pmod{n}$ (clearly divisible by n)

Right side: By the Binomial Theorem:

$$(n+1)^q = \sum_{k=0}^{q} \binom{q}{k} n^k \cdot 1^{q-k} = n^q + \binom{q}{1} n^{q-1} + \ldots + \binom{q}{q-1} n + 1$$

All terms contain n except the last term, so:

$$(n+1)^q \equiv 1 \pmod{n}$$

Step 4: Derive contradiction

From $n^p = (n+1)^q$, we have modulo n:

$$0 \equiv 1 \pmod{n}$$

This means $n \mid (0-1)$, so $n \mid (-1)$.
For $n > 1$, this is impossible.

Conclusion

The assumption that $\log_n(n+1)$ is rational leads to a contradiction.
Therefore, $\log_n(n+1)$ is irrational for all integers $n > 1$.

□

Takeaways 2.15

- **Logarithm to Exponential:** Converting $\log_n(n+1) = \frac{p}{q}$ to $n^p = (n+1)^q$ enables algebraic manipulation
- **Modular Analysis:** Working mod n reveals contradiction: LHS $\equiv 0$ but RHS $\equiv 1$
- **Binomial Expansion:** $(n+1)^q \equiv 1 \pmod{n}$ since only constant term survives
- **Non-standard Irrationality:** Unlike $\sqrt{p}$ proofs, this uses modular arithmetic rather than prime factorization

Problem 2.16: Power Difference Divisible by x Minus y

Prove by induction that $x^n - y^n$ is divisible by $x - y$ for all positive integers n, where x, y are integers with $x \neq y$.

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.16

Proof by Mathematical Induction

Base Case ($n = 1$):

For $n = 1$:

$$x^1 - y^1 = x - y$$

Since $x - y$ is divisible by $x - y$, the statement holds for $n = 1$. ✓

Inductive Hypothesis:

Assume the statement is true for $n = k$, where k is a positive integer:

$$x^k - y^k \text{ is divisible by } x - y$$

This means we can write:

$$x^k - y^k = m(x - y)$$

for some integer m.

Inductive Step:

We must prove the statement for $n = k + 1$:

$$x^{k+1} - y^{k+1} \text{ is divisible by } x - y$$

Start with $x^{k+1} - y^{k+1}$ and manipulate to introduce $x^k - y^k$:

$$\begin{aligned} x^{k+1} - y^{k+1} &= x^{k+1} - xy^k + xy^k - y^{k+1} \\ &= x(x^k - y^k) + y^k(x - y) \end{aligned}$$

Substitute the inductive hypothesis $x^k - y^k = m(x - y)$:

$$\begin{aligned} x^{k+1} - y^{k+1} &= x \cdot m(x - y) + y^k(x - y) \\ &= (x - y)(xm + y^k) \end{aligned}$$

Since x, y, m, k are all integers, $M = xm + y^k$ is an integer.

Therefore:

$$x^{k+1} - y^{k+1} = M(x - y)$$

This shows $x^{k+1} - y^{k+1}$ is divisible by $x - y$. ✓

Conclusion:

By mathematical induction, $x^n - y^n$ is divisible by $x - y$ for all positive integers n.

□

Note: This problem demonstrates the crossover between Induction and Proofs topics in HSC Extension 2.

Takeaways 2.16

- **Strategic Addition/Subtraction:** Add and subtract xy^k to create terms involving $x^k - y^k$ and $x - y$
- **Factorization:** Extract common factor $(x - y)$ after rearrangement
- **Induction Crossover:** Problem appears in both HSC-Induction and HSC-Proofs collections, showing technique overlap
- **Alternative Formula:** Result leads to $x^n - y^n = (x - y)(x^{n-1} + x^{n-2}y + \ldots + xy^{n-2} + y^{n-1})$

Problem 2.17: Divisibility by 6: Criterion and Consequence

For $n \in \mathbb{Z}$, prove that:

(a) n is divisible by 6 if and only if n is divisible by 2 and 3.

(b) $n^3 - n$ is divisible by 6.

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.17

Part (a): Biconditional Proof
Forward ($\Longrightarrow$): If $6 \mid n$, then $2 \mid n$ and $3 \mid n$.

Proof. Assume $6 \mid n$. Then $n = 6k$ for some integer k.
Write $n = 6k = 2(3k)$. Since $3k$ is an integer, $2 \mid n$.
Write $n = 6k = 3(2k)$. Since $2k$ is an integer, $3 \mid n$. □ □

Reverse ($\Longleftarrow$): If $2 \mid n$ and $3 \mid n$, then $6 \mid n$.

Proof. Assume $2 \mid n$ and $3 \mid n$.
Then $n = 2a$ and $n = 3b$ for some integers a, b.
From $n = 2a$, n is even. From $n = 3b$, we have $2a = 3b$.
Since LHS is even, RHS $3b$ must be even. Since 3 is odd, b must be even.
Write $b = 2c$ for some integer c.
Then:

$$n = 3b = 3(2c) = 6c$$

Since c is an integer, $6 \mid n$. □ □

Part (b): Using Part (a)
Goal: Prove $6 \mid (n^3 - n)$.
By part (a), sufficient to show $2 \mid (n^3 - n)$ and $3 \mid (n^3 - n)$.
Step 1: Factor

$$n^3 - n = n(n^2 - 1) = n(n-1)(n+1) = (n-1)n(n+1)$$

This is the product of three consecutive integers.
Step 2: Prove $2 \mid (n^3 - n)$
Among any three consecutive integers, at least one is even.
Therefore, $(n-1)n(n+1)$ contains an even factor, so $2 \mid (n^3 - n)$. ✓
Step 3: Prove $3 \mid (n^3 - n)$
Among any three consecutive integers, exactly one is divisible by 3.
Therefore, $(n-1)n(n+1)$ contains a factor divisible by 3, so $3 \mid (n^3 - n)$. ✓
Conclusion
Since $2 \mid (n^3 - n)$ and $3 \mid (n^3 - n)$, and $\gcd(2, 3) = 1$, by part (a):

$$6 \mid (n^3 - n)$$

□

Takeaways 2.17

- **Coprime Divisibility:** If $\gcd(a, b) = 1$ and both $a \mid n$ and $b \mid n$, then $ab \mid n$
- **Consecutive Integer Properties:** Among k consecutive integers, exactly one is divisible by k
- **Multi-part Strategy:** Part (b) leverages part (a) to simplify proof (check divisibility by 2 and 3 separately)
- **Factorization:** $n^3 - n = (n - 1)n(n + 1)$ reveals structure as consecutive integer product

Problem 2.18: Density of Q in R

Prove that for any real numbers x and y with $x < y$, there exists a rational number q such that

$$x < q < y.$$

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.18

Existence Proof

Since $y - x > 0$, we can choose a positive integer n such that

$$n(y - x) > 1.$$

Equivalently,

$$ny - nx > 1.$$

Now let m be the smallest integer greater than nx. Then

$$m - 1 \leq nx < m.$$

So in particular,

$$nx < m \leq nx + 1.$$

But $ny - nx > 1$ gives

$$nx + 1 < ny.$$

Hence

$$nx < m \leq nx + 1 < ny,$$

so

$$nx < m < ny.$$

Dividing through by $n > 0$,

$$x < \frac{m}{n} < y.$$

Therefore, with $q = \frac{m}{n}$, we have found a rational number satisfying $x < q < y$.

□

Takeaways 2.18

- **Magnification Idea:** Multiply the interval by a large integer so its length becomes greater than 1
- **Key Step:** Once the gap exceeds 1, an integer can be placed between the endpoints
- **Density of $\mathbb{Q}$:** Rational numbers occur between every two distinct real numbers
- **Big Picture:** This is an existence proof rather than a constructive formula for a specific rational
- For any real numbers x and y with $x < y$, there exists infinitely many rational numbers q such that $x < q < y$.

Problem 2.19: Irrationals Between Any Two Rationals

Prove that for any two distinct rational numbers p and q, there exists an irrational number x such that

$$p < x < q$$

or

$$q < x < p.$$

Hint: Attempt the proof independently first. Focus on the key theorem, algebraic transformation, or contradiction setup that links the hypothesis to the target conclusion.

Solution 2.19

Constructive Proof

Without loss of generality, assume $p < q$. Define

$$x = p + \frac{\sqrt{2}}{2}(q-p).$$

Since $0 < \frac{\sqrt{2}}{2} < 1$ and $q - p > 0$, multiplying by $(q-p)$ gives

$$0 < \frac{\sqrt{2}}{2}(q-p) < q - p.$$

Adding p throughout,

$$p < x < q.$$

It remains to show that x is irrational. Suppose, for contradiction, that x is rational. Then $x - p$ is rational, so

$$x - p = \frac{\sqrt{2}}{2}(q-p)$$

is rational. Since $q - p$ is a nonzero rational number, dividing by it gives

$$\frac{\sqrt{2}}{2} = \frac{x-p}{q-p},$$

which is rational. Hence $\sqrt{2}$ would be rational, a contradiction.

Therefore x is irrational, and we have found an irrational number strictly between p and q.

□

Takeaways 2.19

- **Constructive Method:** The proof gives an explicit irrational number between the two rationals
- **Scaling Trick:** Multiply the rational gap $(q - p)$ by an irrational factor between 0 and 1
- **Density of Irrationals:** Irrational numbers occur between any two distinct rationals
- **Companion Result:** Together with density of rationals, this shows the two sets are interwoven on the real line
- For any two distinct rational numbers p and q, there exists infinitely many irrational numbers x such that $p < x < q$ or $q < x < p$.
- Because the two sets ($\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$) of numbers are interwoven, no matter how closely you zoom in on the real line, you will always find both rational and irrational numbers.
- While they are perfectly interwoven and mutually dense, they are not equal. The rationals are countable, meaning the set of all rational numbers can be put into a one-to-one correspondence with the natural numbers, while the irrationals are uncountable, meaning there are "more" irrationals than rationals in a precise mathematical sense.
- The probability of randomly selecting a rational number the interval $[0, 1]$ is zero! Learn "Measure Theory" to understand this concept of "size" of sets in a more rigorous way.

Problem 2.20: Rationality of Cube Roots

(i) Prove that $\sqrt[3]{10}$ is irrational.

(ii) And hence, or otherwise, prove it is not possible for both $\sqrt[3]{2n}$ and $\sqrt[3]{5n}$ to be rational for any positive integer n.

Hint:

- **(i)** Use a proof by contradiction. Assume $\sqrt[3]{10} = \frac{p}{q}$ where p and q are coprime integers. Cube both sides and analyze the parity (divisibility by 2) of both sides to find a contradiction.
- **(ii)** If both are rational, set them equal to rational fractions, cube them, and isolate n. Equate the two expressions for n. Apply the Fundamental Theorem of Arithmetic (specifically looking at the multiplicity of the prime factor 2 or 5) to prove the resulting equation is impossible.

Solution 2.20

(i) Prove that $\sqrt[3]{10}$ is irrational.

Proof by Contradiction

Assume for the sake of contradiction that $\sqrt[3]{10}$ is rational. Then it can be expressed in simplest fractional form as $\frac{p}{q}$, where $p, q \in \mathbb{Z}^+$, $q \neq 0$, and $\gcd(p, q) = 1$.

$$10 = \frac{p^3}{q^3}$$
$$p^3 = 10q^3$$
$$p^3 = 2(5q^3)$$

Since $5q^3$ is an integer, p^3 is a multiple of 2 (it is even). The cube of an odd number is odd, so p must also be even. Let $p = 2k$ for some integer k. Substituting this back:

$$(2k)^3 = 10q^3$$
$$8k^3 = 10q^3$$
$$4k^3 = 5q^3$$

The left side, $4k^3 = 2(2k^3)$, is even. Therefore, the right side, $5q^3$, must also be even. Since 5 is odd, q^3 must be even, which implies q is even.

If both p and q are even, they share a common factor of 2. This contradicts our initial assumption that $\gcd(p, q) = 1$. Therefore, $\sqrt[3]{10}$ is irrational.

□

(ii) Prove it is not possible for both $\sqrt[3]{2n}$ and $\sqrt[3]{5n}$ to be rational.

Proof by Contradiction

Assume for contradiction that for some positive integer n, both expressions are rational. Let $\sqrt[3]{2n} = \frac{a}{b}$ and $\sqrt[3]{5n} = \frac{c}{d}$, where $a, b, c, d \in \mathbb{Z}^+$ and the fractions are in simplest form. Cubing both equations yields:

$$2n = \frac{a^3}{b^3} \implies n = \frac{a^3}{2b^3}$$
$$5n = \frac{c^3}{d^3} \implies n = \frac{c^3}{5d^3}$$

Equating the two expressions for n:

$$\frac{a^3}{2b^3} = \frac{c^3}{5d^3}$$
$$5a^3d^3 = 2b^3c^3$$

Let $X = ad$ and $Y = bc$, where $X, Y \in \mathbb{Z}^+$. Thus, $5X^3 = 2Y^3$.

By the Fundamental Theorem of Arithmetic, we analyze the prime factor 2 on both sides:

- In X^3, the prime 2 appears a multiple of 3 times (say, $3k$ times). Since 5 is odd, the left side $5X^3$ has exactly $3k$ prime factors of 2.
- In Y^3, the prime 2 appears a multiple of 3 times (say, $3m$ times). The coefficient 2 adds one more, meaning the right side $2Y^3$ has exactly $3m + 1$ prime factors of 2.

Equating the multiplicities gives $3k = 3m + 1 \implies 3(k - m) = 1$. Since k and m are integers, their difference cannot be $\frac{1}{3}$. This contradiction proves that both expressions cannot be rational simultaneously.

□

Takeaways 2.20

- **Parity as a Filter:** Part (i) demonstrates that analyzing divisibility (specifically by 2) is a highly efficient mechanism for contradiction proofs involving rational roots.
- **Fundamental Theorem of Arithmetic:** Part (ii) highlights that comparing the exponents of prime factorizations across an equality is a definitive way to prove impossibility in integer equations.

3 Part 2: Problems with Hints and Solutions

The following problems provide strategic hints followed by worked solutions. Attempt each problem independently before consulting the hint, then use the solution to verify your argument structure and mathematical details.

Easy Problems (5 problems)

Problem 3.1: Parity of a Cubed Minus a Plus One

Prove that the expression $a^3 - a + 1$ is odd for all positive integer values of a.

Hint: Consider factoring $a^3 - a$ as a product of consecutive integers. What can you say about the parity of any product of consecutive integers? Alternatively, try proof by cases: analyze when a is even and when a is odd separately.

Solution 3.1

Method 1 (Factorization - Elegant):
Factor the expression:

$$\begin{aligned} a^3 - a + 1 &= a(a^2 - 1) + 1 \\ &= a(a-1)(a+1) + 1 \\ &= (a-1)a(a+1) + 1 \end{aligned}$$

The product $(a-1)a(a+1)$ is the product of three consecutive integers. In any set of consecutive integers, at least one must be even, so the product is even. Let $(a-1)a(a+1) = 2k$ for some integer k. Therefore, $a^3 - a + 1 = 2k + 1$, which is odd by definition.

Method 2 (Cases):
Case 1: If $a = 2m$ (even), then $a^3 - a + 1 = 8m^3 - 2m + 1 = 2(4m^3 - m) + 1$ (odd).
Case 2: If $a = 2m + 1$ (odd), then expanding shows $a^3 - a + 1 = 2(\text{integer}) + 1$ (odd).
In both cases, the expression is odd. □

Takeaways 3.1

- Reconstruct the full proof from the hint and the solution outline, and justify every transformation explicitly.
- Check edge cases and verify where each assumption is used in the argument.

Problem 3.2: Rational Plus Irrational Is Irrational

If a is rational and b is irrational, prove that $a + b$ is irrational.

Hint: Use proof by contradiction. Assume $a + b$ is rational, then solve for b in terms of rational numbers. What property of rational numbers does this contradict?

Solution 3.2

Proof by Contradiction:
Assume, for contradiction, that $a + b$ is rational.
Since a is rational and $a + b$ is rational (by assumption), we can write:

$$a = \frac{p}{q} \quad \text{where } p, q \in \mathbb{Z}, q \neq 0$$
$$a + b = \frac{r}{s} \quad \text{where } r, s \in \mathbb{Z}, s \neq 0$$

Solving for b:

$$b = (a + b) - a = \frac{r}{s} - \frac{p}{q} = \frac{rq - ps}{sq}$$

Since r, q, p, s are all integers and $sq \neq 0$, this shows b is expressible as a ratio of two integers.
Therefore, b must be rational, which **contradicts** the given condition that b is irrational.
Hence, our assumption was false, and $a + b$ must be irrational. $\square$

Takeaways 3.2

- Reconstruct the full proof from the hint and the solution outline, and justify every transformation explicitly.
- Check edge cases and verify where each assumption is used in the argument.

Problem 3.3: An Irrational Power That Is Rational

Prove that an irrational number raised to the power of an irrational number can be rational, by considering $\sqrt{2}^{\sqrt{2}}$. You may assume $\sqrt{2}$ is irrational.

Hint: Consider $\alpha = \sqrt{2}^{\sqrt{2}}$. By the Law of Excluded Middle, α is either rational or irrational. Examine both cases:

- If α is rational, you're done immediately.
- If α is irrational, compute $\alpha^{\sqrt{2}}$ and simplify.

This is a **non-constructive** existence proof—you prove something exists without determining which case actually holds!

Solution 3.3

Proof by Cases:

Let $\alpha = \sqrt{2}^{\sqrt{2}}$. We consider two exhaustive cases:

Case 1: If $\alpha = \sqrt{2}^{\sqrt{2}}$ is rational, then we have found irrational base $\sqrt{2}$ and irrational exponent $\sqrt{2}$ such that $\sqrt{2}^{\sqrt{2}}$ is rational. Done.

Case 2: If $\alpha = \sqrt{2}^{\sqrt{2}}$ is irrational, consider:

$$\begin{aligned}
\alpha^{\sqrt{2}} &= \left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} \\
&= \sqrt{2}^{\sqrt{2}\cdot\sqrt{2}} \quad \text{(exponent law)} \\
&= \sqrt{2}^{2} \\
&= 2
\end{aligned}$$

Since 2 is rational and both α (irrational by case assumption) and $\sqrt{2}$ (given as irrational) are irrational, we have found an example.

Conclusion: In either case, there exist irrational numbers a and b such that a^b is rational. $\square$

Note: This proof doesn't tell us whether $\sqrt{2}^{\sqrt{2}}$ is actually rational or irrational—and we don't need to know! This is the beauty of non-constructive existence proofs.

Takeaways 3.3

- Reconstruct the full proof from the hint and the solution outline, and justify every transformation explicitly.
- Check edge cases and verify where each assumption is used in the argument.

Problem 3.4: Consecutive Divisibility Is Impossible

Prove that for positive integers a, b with $a > 1$, either b is not divisible by a or $b + 1$ is not divisible by a (or both).

Hint: Use proof by contradiction. Assume both $a|b$ and $a|(b+1)$. What does this tell you about $a|[(b+1) - b]$? What can you conclude about a?

Solution 3.4

Proof by Contradiction:
The statement is equivalent to: "It is not the case that both $a|b$ and $a|(b+1)$."
Assume, for contradiction, that **both** $a|b$ and $a|(b+1)$.
Then there exist integers k_1 and k_2 such that:

$$b = ak_1$$
$$b + 1 = ak_2$$

Subtracting the first equation from the second:

$$(b+1) - b = ak_2 - ak_1$$

$$1 = a(k_2 - k_1)$$

Let $K = k_2 - k_1 \in \mathbb{Z}$. Then $1 = aK$, which means a divides 1.
Since a is a positive integer, the only positive divisor of 1 is 1 itself. Therefore, $a = 1$.
This **contradicts** the given condition that $a > 1$.
Hence, our assumption must be false, and at least one of b or $b + 1$ is not divisible by a. □

Takeaways 3.4

- Reconstruct the full proof from the hint and the solution outline, and justify every transformation explicitly.
- Check edge cases and verify where each assumption is used in the argument.

Problem 3.5: Mersenne Prime Exponent Cannot Be Even

Prove that for all integers $n \geq 3$, if $2^n - 1$ is prime, then n cannot be even.

Hint: Use proof by contrapositive. Instead of proving "$P \implies Q$", prove "$\neg Q \implies \neg P$".
That is, prove: "If n is even (and $n \geq 3$), then $2^n - 1$ is composite."
Write $n = 2k$ and factor $2^{2k} - 1$ as a difference of squares.

Solution 3.5

Proof by Contrapositive:
We prove the contrapositive: If n is even (with $n \geq 3$), then $2^n - 1$ is composite.
Assume n is even. Then $n = 2k$ for some integer k.
Since $n \geq 3$ and n is even, we have $n \geq 4$, so $k \geq 2$.
Substitute:

$$2^n - 1 = 2^{2k} - 1 = (2^k)^2 - 1^2$$

Factor as difference of squares:

$$2^n - 1 = (2^k - 1)(2^k + 1)$$

Since $k \geq 2$:

- $2^k \geq 4$, so $2^k - 1 \geq 3 > 1$
- $2^k \geq 4$, so $2^k + 1 \geq 5 > 1$

Both factors are integers strictly greater than 1, so their product is composite.
Therefore, $2^n - 1$ is composite.

By contrapositive, if $2^n - 1$ is prime, then n cannot be even. $\square$

Note: This is why Mersenne primes have the form $2^p - 1$ where p itself is prime (though not all such numbers are prime).

Takeaways 3.5

- Reconstruct the full proof from the hint and the solution outline, and justify every transformation explicitly.
- Check edge cases and verify where each assumption is used in the argument.

Medium Problems (5 problems)

Problem 3.6: Divisibility of a Shifted Power Expression

Prove that $(2^m - 1)^2 - 1$ is divisible by 2^{m+1} for all integers $m \geq 1$.

Hint: Expand the perfect square $(2^m - 1)^2$ algebraically, then simplify. Try to factor out 2^{m+1} from the resulting expression.

Solution 3.6

Let $E = (2^m - 1)^2 - 1$.
Step 1: Expand the square:

$$\begin{aligned} E &= (2^m)^2 - 2(2^m) + 1 - 1 \\ &= 2^{2m} - 2^{m+1} + 1 - 1 \\ &= 2^{2m} - 2^{m+1} \end{aligned}$$

Step 2: Factor out 2^{m+1}:
Note that $2^{2m} = 2^{m+1} \cdot 2^{m-1}$ (using exponent laws).
Therefore:

$$\begin{aligned} E &= 2^{m+1} \cdot 2^{m-1} - 2^{m+1} \cdot 1 \\ &= 2^{m+1}(2^{m-1} - 1) \end{aligned}$$

Step 3: Conclude divisibility:
Since $m \geq 1$, we have $m - 1 \geq 0$, so 2^{m-1} is an integer. Thus $(2^{m-1} - 1)$ is an integer.
Therefore, $E = 2^{m+1} \cdot k$ where $k = 2^{m-1} - 1 \in \mathbb{Z}$.
By definition, $(2^m - 1)^2 - 1$ is divisible by 2^{m+1}. $\square$

Takeaways 3.6

- Reconstruct the full proof from the hint and the solution outline, and justify every transformation explicitly.
- Check edge cases and verify where each assumption is used in the argument.

Problem 3.7: Difference of Squares Cannot Equal 1

Prove that there are no positive integers x, y such that $x^2 - y^2 = 1$.

Hint: Factor the left side as a difference of squares: $(x-y)(x+y) = 1$. For this product to equal 1, what must be true about the integer factors $(x-y)$ and $(x+y)$? Consider all possible integer factor pairs of 1.

Solution 3.7

Factor the equation:

$$x^2 - y^2 = 1 \implies (x-y)(x+y) = 1$$

Since x and y are integers, both $(x-y)$ and $(x+y)$ are integers.
For the product of two integers to equal 1, the only possibilities are:

- $x - y = 1$ and $x + y = 1$, or
- $x - y = -1$ and $x + y = -1$

Case 1: $x - y = 1$ and $x + y = 1$
Adding these equations: $2x = 2$, so $x = 1$.
Substituting back: $1 + y = 1$, so $y = 0$.
Since $y = 0$ is not a positive integer, this case fails.

Case 2: $x - y = -1$ and $x + y = -1$
Adding these equations: $2x = -2$, so $x = -1$.
Since $x = -1$ is not a positive integer, this case fails.

Conclusion: Neither case yields a solution with both x and y positive integers.
Therefore, there are no positive integers x, y such that $x^2 - y^2 = 1$. $\square$

Takeaways 3.7

- Reconstruct the full proof from the hint and the solution outline, and justify every transformation explicitly.
- Check edge cases and verify where each assumption is used in the argument.

Problem 3.8: Divisibility of a Product

Prove that for all positive integers a, b, and c, if a divides the product bc, then a divides b or a divides c.

Hint: Attempt to set up a direct proof using the definition of divisibility: $bc = ak$ for some integer k. Can you algebraically force a to divide b or c?
Wait—is this statement actually true? Try testing it with some small integer values. What happens if a is a composite number (like 4 or 6) rather than a prime number?
Exam Strategy: If you ever encounter an exam question that seems fundamentally incorrect or impossible, do not leave the page blank! A blank page guarantees 0 marks. Instead, prove that the statement is false. The absolute best way to completely destroy a "for all" statement is to provide a single, concrete **counterexample**.

Solution 3.8

The problem statement is strictly **false**. We will disprove it by providing a counterexample.
To disprove the statement "if a divides bc, then $a|b$ or $a|c$", we must find specific positive integers a, b, and c such that a divides bc, but $a \nmid b$ and $a \nmid c$.
Let $a = 6$, $b = 2$, and $c = 3$.
Check the condition (the premise):

$$bc = 2 \times 3 = 6$$

Since 6 divides 6, the premise that a divides bc is satisfied.
Check the conclusion:

- Does a divide b? No, 6 does not divide 2.
- Does a divide c? No, 6 does not divide 3.

Since both parts of the conclusion are false, the entire statement is invalid.
Therefore, it is not true for all positive integers a, b, c that $a|bc \implies a|b$ or $a|c$. $\square$

Takeaways 3.8

- **Trust Your Math, Not the Paper:** Examiners are human and occasionally make mistakes. If you leave a flawed question blank, you receive 0 marks. If you mathematically prove that the question is flawed, you demonstrate mastery and will be awarded marks.
- **The Power of Counterexamples:** To prove a universal statement ("for all $x\ldots$") you must prove it algebraically for every number in existence. But to *disprove* it, you only need to find **one** single case where it fails.
- **Euclid's Lemma:** The statement in this problem becomes completely true if we add one crucial condition: a must be a *prime* number. Composite numbers can have their factors split across b and c, which is why $a = 6$ failed above.

Problem 3.9: Extreme Arguments of a Complex Disk

Consider the region $\mathcal{R}$ in the complex plane defined by $|z - (2\sqrt{3} + 2i)| \leq 2$.

(a) Describe the region $\mathcal{R}$ geometrically (what shape? center? radius?).

(b) Find the maximum and minimum values of $\arg(z)$ for $z \in \mathcal{R}$, where $-\pi < \arg(z) \leq \pi$.

(c) Find the complex number z associated with the maximum argument in part (b) in the form $x + iy$.

Hint:(a) The inequality $|z - w| \leq r$ represents a disk (filled circle) in the complex plane.
(b) Find $\arg(C)$ where $C = 2\sqrt{3} + 2i$ is the center. The max/min arguments occur at tangent lines from the origin to the circle. Use the right triangle formed by the origin, center, and tangent point.
(c) The point with maximum argument lies on the ray from O with angle $\arg_{\max}$ found in (b), at distance $\sqrt{|OC|^2 - r^2}$ from origin.

Solution 3.9

(a) Geometric Description:
The region is a closed disk (circle plus interior) with:

- Center: $C = 2\sqrt{3} + 2i$ (which is $(2\sqrt{3}, 2)$ in Cartesian coordinates)
- Radius: $r = 2$

Distance from origin to center: $|OC| = \sqrt{(2\sqrt{3})^2 + 2^2} = \sqrt{12+4} = 4$.
Since $|OC| = 4 > r = 2$, the origin lies outside the circle.

(b) Max and Min Arguments:
The argument of the center is:

$$\arg(C) = \arctan\left(\frac{2}{2\sqrt{3}}\right) = \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

The tangent lines from O to the circle create a right triangle with:

- Hypotenuse: $|OC| = 4$
- Opposite side (radius): $r = 2$

The angle θ from OC to the tangent satisfies:

$$\sin\theta = \frac{r}{|OC|} = \frac{2}{4} = \frac{1}{2} \implies \theta = \frac{\pi}{6}$$

Therefore:

$$\begin{aligned} \arg_{\max} &= \frac{\pi}{6} + \frac{\pi}{6} = \frac{\pi}{3} \\ \arg_{\min} &= \frac{\pi}{6} - \frac{\pi}{6} = 0 \end{aligned}$$

(c) Complex Number for Max Argument:
The tangent point distance from origin:

$$|OP| = \sqrt{|OC|^2 - r^2} = \sqrt{16-4} = \sqrt{12} = 2\sqrt{3}$$

The point lies on the ray with argument $\pi/3$:

$$\begin{aligned} z &= 2\sqrt{3} \cdot e^{i\pi/3} = 2\sqrt{3}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) \\ &= 2\sqrt{3}\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) \\ &= \sqrt{3} + 3i \end{aligned}$$

Answer: $z = \sqrt{3} + 3i$. □

Takeaways 3.9

- Reconstruct the full proof from the hint and the solution outline, and justify every transformation explicitly.
- Check edge cases and verify where each assumption is used in the argument.

Problem 3.10: Evenness Equivalence for x and x Squared

Prove that x is even if and only if x^2 is even (for integer x).

Hint: This is a biconditional (iff) statement requiring two proofs:

- Direction 1: If x is even, then x^2 is even (direct proof).
- Direction 2: If x^2 is even, then x is even (try contrapositive: if x is odd, then x^2 is odd).

Solution 3.10

Direction 1 ($\Rightarrow$): If x is even, then x^2 is even.
Assume x is even. Then $x = 2k$ for some integer k.
Therefore:

$$x^2 = (2k)^2 = 4k^2 = 2(2k^2)$$

Since $2k^2$ is an integer, x^2 is even by definition.

Direction 2 ($\Leftarrow$): If x^2 is even, then x is even.
We prove the contrapositive: If x is odd, then x^2 is odd.
Assume x is odd. Then $x = 2k + 1$ for some integer k.
Therefore:

$$\begin{aligned} x^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \end{aligned}$$

Since $2k^2 + 2k$ is an integer, x^2 is odd by definition.
By contrapositive, if x^2 is even, then x must be even.

Conclusion: Both directions proven, so x is even $\iff$ x^2 is even. $\square$

Note: This result is fundamental in many irrationality proofs (e.g., $\sqrt{2}$ is irrational). If p is an odd prime and $p|x^2$, then $p|x$.

Takeaways 3.10

- Reconstruct the full proof from the hint and the solution outline, and justify every transformation explicitly.
- Check edge cases and verify where each assumption is used in the argument.

Hard Problems (6 problems)

Problem 3.11: Product of Two Irrationals: Counterexample

Prove or disprove: If x and y are irrational numbers with $x \neq y$, then xy is irrational.

Hint: The statement is **false**. Find a counterexample using multiples of $\sqrt{2}$. Consider $x = \sqrt{2}$ and $y = k\sqrt{2}$ for some rational $k \neq 1$. What is xy? Is it rational or irrational?

Solution 3.11

The statement is **false**. We disprove by counterexample.

Counterexample:
Let $x = \sqrt{2}$ and $y = 2\sqrt{2}$.

- **Check x is irrational:** $\sqrt{2}$ is irrational (well-known).
- **Check y is irrational:** $2\sqrt{2}$ is the product of rational 2 and irrational $\sqrt{2}$, so it's irrational.
- **Check $x \neq y$:** Clearly $\sqrt{2} \neq 2\sqrt{2}$ since $1 \neq 2$.
- **Compute xy:**
$$xy = (\sqrt{2})(2\sqrt{2}) = 2 \cdot (\sqrt{2})^2 = 2 \cdot 2 = 4$$
- **Check rationality of xy:** $4 = \frac{4}{1}$ is **rational**.

Since we found irrational numbers x and y with $x \neq y$ such that xy is rational, the original statement is **disproven**. □

Note: This shows that the irrationals are not closed under multiplication. The rationals are closed under multiplication, but the irrationals are not.

Takeaways 3.11

- Reconstruct the full proof from the hint and the solution outline, and justify every transformation explicitly.
- Check edge cases and verify where each assumption is used in the argument.

Problem 3.12: Parity Obstruction in Pythagorean Triples

Prove that there is no Pythagorean triple (a, b, c) where a and b (the two smallest numbers) are both even and c (the largest number) is odd.

Hint: Use proof by contradiction. Assume such a triple exists with $a = 2k$ and $b = 2m$ (both even) and c odd. Substitute into the Pythagorean equation $a^2 + b^2 = c^2$ and analyze the parity of c^2. What can you conclude about the parity of c?

Solution 3.12

Proof by Contradiction:
Assume there exists a Pythagorean triple (a, b, c) where:

- $a^2 + b^2 = c^2$
- a and b are both even
- c is odd

Since a and b are even, write $a = 2k$ and $b = 2m$ for integers k, m.

Substitute into Pythagorean equation:

$$(2k)^2 + (2m)^2 = c^2$$
$$4k^2 + 4m^2 = c^2$$
$$4(k^2 + m^2) = c^2$$

Analyze parity of c^2:
The equation shows $c^2 = 4(k^2 + m^2)$, which is a multiple of 4.
In particular, c^2 is divisible by 2, so c^2 is **even**.

Derive parity of c:
If c were odd, then $c = 2n + 1$ for some integer n, and:

$$c^2 = (2n + 1)^2 = 4n^2 + 4n + 1 = 2(2n^2 + 2n) + 1$$

This shows c^2 would be **odd**, contradicting that c^2 is even.
Therefore, c must be **even**.

Contradiction:
We derived that c must be even, which contradicts our assumption that c is odd.
Hence, no such Pythagorean triple exists. $\square$

Takeaways 3.12

- Reconstruct the full proof from the hint and the solution outline, and justify every transformation explicitly.
- Check edge cases and verify where each assumption is used in the argument.

Problem 3.13: Exponential Comparison of Three Four and Five

Prove or disprove: There exists a real number n such that $3^n + 4^n < 5^n$.

Hint: The statement is **true**. Try a few values: $n = 2$ gives $9 + 16 = 25 = 5^2$ (equality, not $<$).
Try $n = 3$: Is $3^3 + 4^3 < 5^3$? Calculate $27 + 64$ vs 125.
Alternatively, divide by 5^n to get $(3/5)^n + (4/5)^n < 1$ and analyze the function behavior.

Solution 3.13

The statement is **true**.

Proof by Example:
Try $n = 3$:

$$3^3 + 4^3 = 27 + 64 = 91$$
$$5^3 = 125$$

Since $91 < 125$, the inequality $3^3 + 4^3 < 5^3$ holds.
Therefore, $n = 3$ is a real number satisfying the inequality. $\square$

Alternative Proof (Analysis):
Divide the inequality by 5^n:

$$\left(\frac{3}{5}\right)^n + \left(\frac{4}{5}\right)^n < 1$$

Let $f(n) = \left(\frac{3}{5}\right)^n + \left(\frac{4}{5}\right)^n$.

- At $n = 2$: $f(2) = \frac{9}{25} + \frac{16}{25} = 1$ (equality)
- The derivative: $f'(n) = \left(\frac{3}{5}\right)^n \ln\left(\frac{3}{5}\right) + \left(\frac{4}{5}\right)^n \ln\left(\frac{4}{5}\right)$
  Since $\frac{3}{5} < 1$ and $\frac{4}{5} < 1$, both logarithms are negative, so $f'(n) < 0$.
- Therefore, $f(n)$ is strictly decreasing.

Since $f(2) = 1$ and f is strictly decreasing, for any $n > 2$, we have $f(n) < 1$.
Thus, the inequality holds for all $n > 2$. In particular, it holds for $n = 3$ (and infinitely many other values). $\square$

Note: This shows the power of the "divide by largest term" technique in inequality problems. The critical point is $n = 2$ where equality holds (Pythagorean triple: $3^2 + 4^2 = 5^2$).

Takeaways 3.13

- Reconstruct the full proof from the hint and the solution outline, and justify every transformation explicitly.
- Check edge cases and verify where each assumption is used in the argument.

Problem 3.14: Cubic Sums Bound a Quartic Term

Prove by induction that for all integers $n \geq 2$:

$$1^3 + 2^3 + \cdots + (n-1)^3 < \frac{n^4}{4} < 1^3 + 2^3 + \cdots + n^3$$

Hint: This is a "sandwich" inequality with two parts:

- LHS inequality: $\sum_{k=1}^{n-1} k^3 < \frac{n^4}{4}$ (needs induction)
- RHS inequality: $\frac{n^4}{4} < \sum_{k=1}^{n} k^3$ (can be proven directly)

Recall: $\sum_{k=1}^{n} k^3 = \left[\frac{n(n+1)}{2}\right]^2 = \frac{n^2(n+1)^2}{4}$

For the LHS induction step, you'll need to show $\frac{k^4}{4} + k^3 < \frac{(k+1)^4}{4}$, which simplifies to $0 < 6k^2 + 4k + 1$.

Solution 3.14

Part 1: RHS Inequality (Direct Proof)
Show $\frac{n^4}{4} < \sum_{k=1}^{n} k^3$ for $n \geq 2$.
Using the sum formula:

$$\sum_{k=1}^{n} k^3 = \frac{n^2(n+1)^2}{4} = \frac{n^4 + 2n^3 + n^2}{4}$$

We need: $\frac{n^4}{4} < \frac{n^4+2n^3+n^2}{4}$
This simplifies to: $0 < 2n^3 + n^2 = n^2(2n+1)$
Since $n \geq 2 > 0$, this is clearly true. RHS proven. ✓

Part 2: LHS Inequality (Induction)
Prove $\sum_{k=1}^{n-1} k^3 < \frac{n^4}{4}$ for $n \geq 2$.

Base case ($n = 2$):

$$\sum_{k=1}^{1} k^3 = 1^3 = 1 < \frac{2^4}{4} = 4 \quad \checkmark$$

Inductive hypothesis: Assume $\sum_{k=1}^{m-1} k^3 < \frac{m^4}{4}$ for some $m \geq 2$.

Inductive step: Prove $\sum_{k=1}^{m} k^3 < \frac{(m+1)^4}{4}$.

$$\begin{aligned} \sum_{k=1}^{m} k^3 &= \sum_{k=1}^{m-1} k^3 + m^3 \\ &< \frac{m^4}{4} + m^3 \quad \text{(by IH)} \\ &= \frac{m^4 + 4m^3}{4} \end{aligned}$$

Need to show: $\frac{m^4+4m^3}{4} < \frac{(m+1)^4}{4}$
Equivalently: $m^4 + 4m^3 < (m+1)^4$
Expand RHS:

$$(m+1)^4 = m^4 + 4m^3 + 6m^2 + 4m + 1$$

So we need: $m^4 + 4m^3 < m^4 + 4m^3 + 6m^2 + 4m + 1$
Simplifies to: $0 < 6m^2 + 4m + 1$
This is true for all $m \geq 2$. ✓

By induction, LHS proven. Combining both parts, the full sandwich inequality holds. □

Takeaways 3.14

- Reconstruct the full proof from the hint and the solution outline, and justify every transformation explicitly.
- Check edge cases and verify where each assumption is used in the argument.

Problem 3.15: Induction for a Divisibility Pattern Modulo Ten

Prove by induction that $4^{n+1} + 6^n$ is divisible by 10 when n is an even positive integer.

Hint: Since n is even, let $n = 2k$ for $k \geq 1$. Prove by induction on k that $4^{2k+1} + 6^{2k}$ is divisible by 10.

For the inductive step:

- Express $4^{2(k+1)+1} + 6^{2(k+1)}$ as $16 \cdot 4^{2k+1} + 36 \cdot 6^{2k}$
- From IH: $6^{2k} = 10M - 4^{2k+1}$ for some integer M
- Substitute and factor out 10

Solution 3.15

Since n is an even positive integer, let $n = 2k$ where $k \in \mathbb{Z}^+$.
We prove by induction on k that $P(k)$: "$4^{2k+1} + 6^{2k}$ is divisible by 10" holds for all $k \geq 1$.

Base case ($k = 1$, i.e., $n = 2$):

$$4^{2(1)+1} + 6^{2(1)} = 4^3 + 6^2 = 64 + 36 = 100 = 10 \times 10$$

Divisible by 10. ✓

Inductive hypothesis:
Assume $P(k)$ holds: $4^{2k+1} + 6^{2k} = 10M$ for some integer M.
From this: $6^{2k} = 10M - 4^{2k+1}$

Inductive step:
Prove $P(k+1)$: $4^{2(k+1)+1} + 6^{2(k+1)}$ is divisible by 10.

$$\begin{aligned} 4^{2(k+1)+1} + 6^{2(k+1)} &= 4^{2k+3} + 6^{2k+2} \\ &= 4^2 \cdot 4^{2k+1} + 6^2 \cdot 6^{2k} \\ &= 16 \cdot 4^{2k+1} + 36 \cdot 6^{2k} \end{aligned}$$

Substitute $6^{2k} = 10M - 4^{2k+1}$ from IH:

$$\begin{aligned} &= 16 \cdot 4^{2k+1} + 36(10M - 4^{2k+1}) \\ &= 16 \cdot 4^{2k+1} + 360M - 36 \cdot 4^{2k+1} \\ &= 360M + (16 - 36) \cdot 4^{2k+1} \\ &= 360M - 20 \cdot 4^{2k+1} \\ &= 10(36M - 2 \cdot 4^{2k+1}) \end{aligned}$$

Since M and 4^{2k+1} are integers, $36M - 2 \cdot 4^{2k+1}$ is an integer.
Therefore, $4^{2(k+1)+1} + 6^{2(k+1)}$ is divisible by 10. ✓

By the principle of mathematical induction, $P(k)$ holds for all $k \geq 1$.
Therefore, $4^{n+1} + 6^n$ is divisible by 10 for all even positive integers n. $\square$

Takeaways 3.15

- Reconstruct the full proof from the hint and the solution outline, and justify every transformation explicitly.
- Check edge cases and verify where each assumption is used in the argument.

Problem 3.16: Leading Digits of Powers of Two

The Pigeonhole Principle and Leading Digits

For any real number x, the fractional part of x is denoted by $\{x\}$ and is defined as $\{x\} = x - \lfloor x \rfloor$, where $\lfloor x \rfloor$ is the greatest integer less than or equal to x. Note that $0 \le \{x\} < 1$.

For example, $\{3.14\} = 0.14$, $\{2026\} = 0$.

(i) Show that for any positive integer M, the ordinary decimal representation of 2^M begins with the digits 2025 (i.e., $2^M = 2025\ldots$) if and only if:

$$\log_{10} 2.025 \le \{M \log_{10} 2\} < \log_{10} 2.026$$

(ii) Let α be an irrational number. Prove that for any two distinct integers m and n, the fractional parts $\{m\alpha\}$ and $\{n\alpha\}$ must be strictly distinct.

(iii) Let N be a positive integer. By dividing the interval $[0, 1)$ into N equal subintervals, use the Pigeonhole Principle to prove that there exists a positive integer $k \le N$ such that:

$$0 < \{k\alpha\} < \frac{1}{N}$$

(iv) Given that $\alpha = \log_{10} 2$ is an irrational number, let

$$\delta = \log_{10} 2.026 - \log_{10} 2.025.$$

By choosing an integer N such that $\frac{1}{N} < \delta$, or otherwise, deduce that there exists at least one positive integer M such that 2^M begins with the digits 2025.

Hint:

- **(i)** Write 2^M in scientific notation: $2^M = c \times 10^K$. If the number starts with 2025, what is c bounded between? Take the base-10 logarithm.
- **(ii)** Proof by contradiction. Assume $\{m\alpha\} = \{n\alpha\}$. Expand using the definition $\{x\} = x - \lfloor x \rfloor$ and isolate α.
- **(iii)** Consider the $N + 1$ fractional parts $\{0\}, \{\alpha\}, \{2\alpha\}, \ldots, \{N\alpha\}$. Place these „pigeons" into the N „holes" (subintervals).
- **(iv)** Think of the interval as a target of width δ. Part (iii) guarantees a „step size" $\{k\alpha\}$ strictly smaller than δ. Can consecutive steps jump over the target without landing inside it?

Solution 3.16

Part (i): 2^M begins with 2025 if and only if $2^M = c \times 10^K$ for some integer $K \geq 0$ and $2.025 \leq c < 2.026$. Taking logarithms yields $M \log_{10} 2 = K + \log_{10} c$. Since $0 \leq \log_{10} c < 1$, we must have $K = \lfloor M \log_{10} 2 \rfloor$ and $\log_{10} c = \{M \log_{10} 2\}$. Substituting this into the inequality for c yields the required result.

Part (ii): Assume $\{m\alpha\} = \{n\alpha\}$ for integers $m \neq n$. Then $m\alpha - \lfloor m\alpha \rfloor = n\alpha - \lfloor n\alpha \rfloor$, which implies

$$\alpha = \frac{\lfloor m\alpha \rfloor - \lfloor n\alpha \rfloor}{m - n}.$$

Since the numerator is an integer and the denominator is a non-zero integer, this forces α to be rational, contradicting the premise. Thus, the fractional parts are distinct.

Part (iii): Divide $[0, 1)$ into N equal intervals of width $1/N$. By (ii), the $N+1$ values $\{0\}, \{\alpha\}, \ldots, \{N\alpha\}$ are strictly distinct. By the Pigeonhole Principle, two values $\{i\alpha\} > \{j\alpha\}$ must fall in the same interval. Thus, $0 < \{i\alpha\} - \{j\alpha\} < 1/N$. Since

$$\{i\alpha\} - \{j\alpha\} = (i - j)\alpha - (\lfloor i\alpha \rfloor - \lfloor j\alpha \rfloor)$$

lies in $(0, 1)$, it is exactly $\{(i - j)\alpha\}$. Let $k = i - j$ (where $0 < k \leq N$) to obtain $0 < \{k\alpha\} < 1/N$.

Part (iv): Choose N such that $1/N < \delta$. From (iii), let our step size be $s = \{k\alpha\}$, noting $0 < s < \delta$. Consider the sequence of multiples $s, 2s, 3s, \ldots$. Because s is strictly smaller than the target interval width δ, a multiple ps must eventually land inside $[\log_{10} 2.025, \log_{10} 2.026) \subset [0, 1)$. Therefore, $\{pk\alpha\} = ps$ satisfies the bounds. By letting $M = pk$, we satisfy the condition from (i).

Takeaways 3.16

- **Kronecker's Theorem:** This problem is a guided proof demonstrating that the orbit of an irrational rotation is dense. It guarantees that by doubling a number enough times, you can eventually generate a power of 2 that starts with any sequence of digits you desire.
- **The Continuous Pigeonhole:** By dividing a continuous number line into physical subintervals, we can force infinite irrational sequences to behave predictably.

Additional Problem

Problem 3.17: Divisibility by 9 via Digit Sum

Prove that a three-digit number is divisible by 9 if and only if the sum of its digits is divisible by 9.

Hint: Let $N = 100a + 10b + c$ where a, b, c are the digits. Rewrite this as $N = 9(11a + b) + (a + b + c)$ to establish the relationship between N and the digit sum. Then prove both directions of the biconditional using this relationship.

Solution 3.17

Let $N = 100a + 10b + c$ be a three-digit number with digits a, b, c.
Let $S = a + b + c$ be the sum of digits.

Key Relationship:
Rewrite N:

$$\begin{aligned} N &= 100a + 10b + c \\ &= (99a + a) + (9b + b) + c \\ &= 9(11a + b) + (a + b + c) \\ &= 9(11a + b) + S \end{aligned}$$

Therefore: $N - S = 9(11a + b)$, which means $N \equiv S \pmod{9}$.

Direction 1 ($\Rightarrow$): If $9|N$, then $9|S$.
If $N \equiv 0 \pmod{9}$, then from $N \equiv S \pmod{9}$, we get $S \equiv 0 \pmod{9}$.
Thus $9|S$.

Direction 2 ($\Leftarrow$): If $9|S$, then $9|N$.
If $S \equiv 0 \pmod{9}$, then from $N \equiv S \pmod{9}$, we get $N \equiv 0 \pmod{9}$.
Thus $9|N$.

Both directions proven, so the biconditional holds. $\square$

Note: This proof generalizes to all positive integers and divisibility by 9. The key is the modular relationship $N \equiv S \pmod{9}$.

Takeaways 3.17

- Reconstruct the full proof from the hint and the solution outline, and justify every transformation explicitly.
- Check edge cases and verify where each assumption is used in the argument.

4 Conclusion

5 Appendix

5.1 De Morgan's Laws

Formal logic is now part of the HSC Extension 2 syllabus. **De Morgan's Laws** tell you how to push a negation ($\neg$) through brackets when a statement uses **AND** ($\wedge$) or **OR** ($\vee$). Get this wrong and a proof by contradiction starts from the wrong assumption.

The laws. For statements P and Q:

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$
$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

Memory trick: distribute $\neg$ inside the brackets and *flip* $\wedge$ to $\vee$ (or $\vee$ to $\wedge$).

Negation of an implication (often tested with contradiction):

$$\neg(P \implies Q) \equiv P \wedge \neg Q$$

The negation of "if P then Q" is *not* another if-then. It is: P happens, but Q fails.

Example 1 (negating OR). "$a \leq 2$ or $b \leq 2$" has negation "$a > 2$ **and** $b > 2$"—Law 2 with P: $a \leq 2$, Q: $b \leq 2$. This is exactly the opening move in Part 1, Problem 8.

Example 2 (contradiction). Prove: if x^2 is even, then x is even ($P \implies Q$).
Assume, for contradiction, the *exact* negation:

$$x^2 \text{ is even} \quad \wedge \quad x \text{ is odd}$$

Write $x = 2k + 1$. Then $x^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, which is odd—contradicting x^2 even. Without $\neg(P \implies Q) \equiv P \wedge \neg Q$, many students incorrectly assume "x^2 even and x even" or invent an invalid if-then negation.

Why this matters. Rigorous proof is not only algebra; it is *precise logic.* Contradiction, contrapositive, and quantified statements all depend on correct negation. University mathematics (analysis, algebra, computer science) treats these equivalences as standard tools—not optional extras. Mastering De Morgan's Laws now prevents silent logical errors later.

Contact Information:

LinkedIn: https://www.linkedin.com/in/nguyenvuhung/

GitHub: https://github.com/vuhung16au/

Repository: https://github.com/vuhung16au/math-olympiad-ml/tree/main/HSC-Proofs